	SR UNIVERSITY Campus Warangal
	Program: II - B. Tech (CS& AI) Batch:
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	Department: Computer Science and AI Semester: I
	AI Assisted Coding-Assignment Date:

Q1. Class with Conditional Logic – Employee Salary

Scenario:
An AI coding assistant is used to develop a simple employee salary management program.

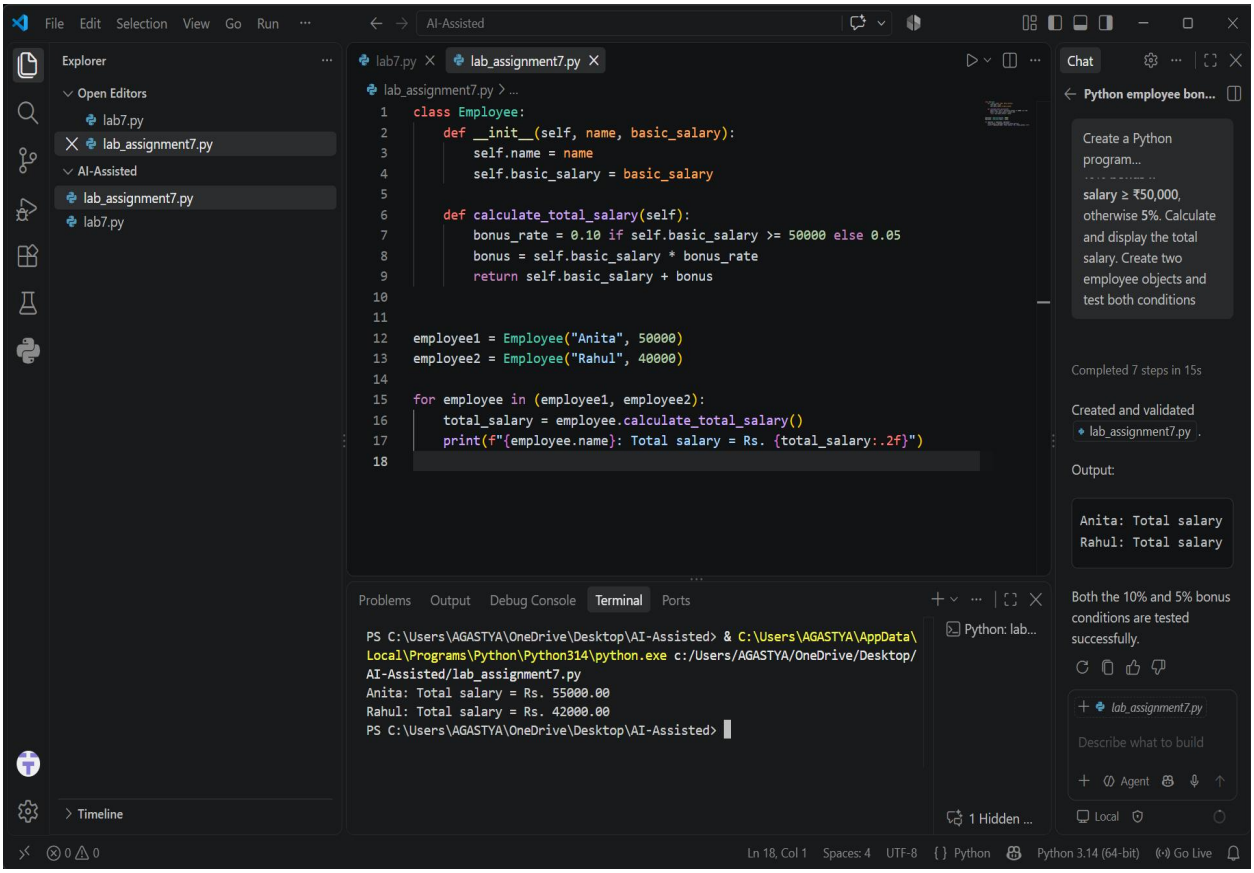
Task:
Using AI-assisted code completion, write a Python class Employee that:

- Stores employee name and basic salary.
- Has a method calculate_bonus() that:
  - Returns 10% of salary if salary is ₹50,000 or above.
  - Returns 5% of salary otherwise.
- Has a method display_details() to display the employee name, salary, and bonus.
- Create at least two employee objects and test the class.

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Prompt:
Create a Python program using an Employee class with name and basic_salary. Calculate a **10% bonus if salary ≥ ₹50,000**, otherwise **5%**. Calculate and display the total salary. Create two employee objects and test both conditions

Code & output:



Observation:

Learned to create classes and objects, use methods, and apply conditional logic to calculate employee bonuses.

Q2. Loop and Conditional Logic – Student Marks

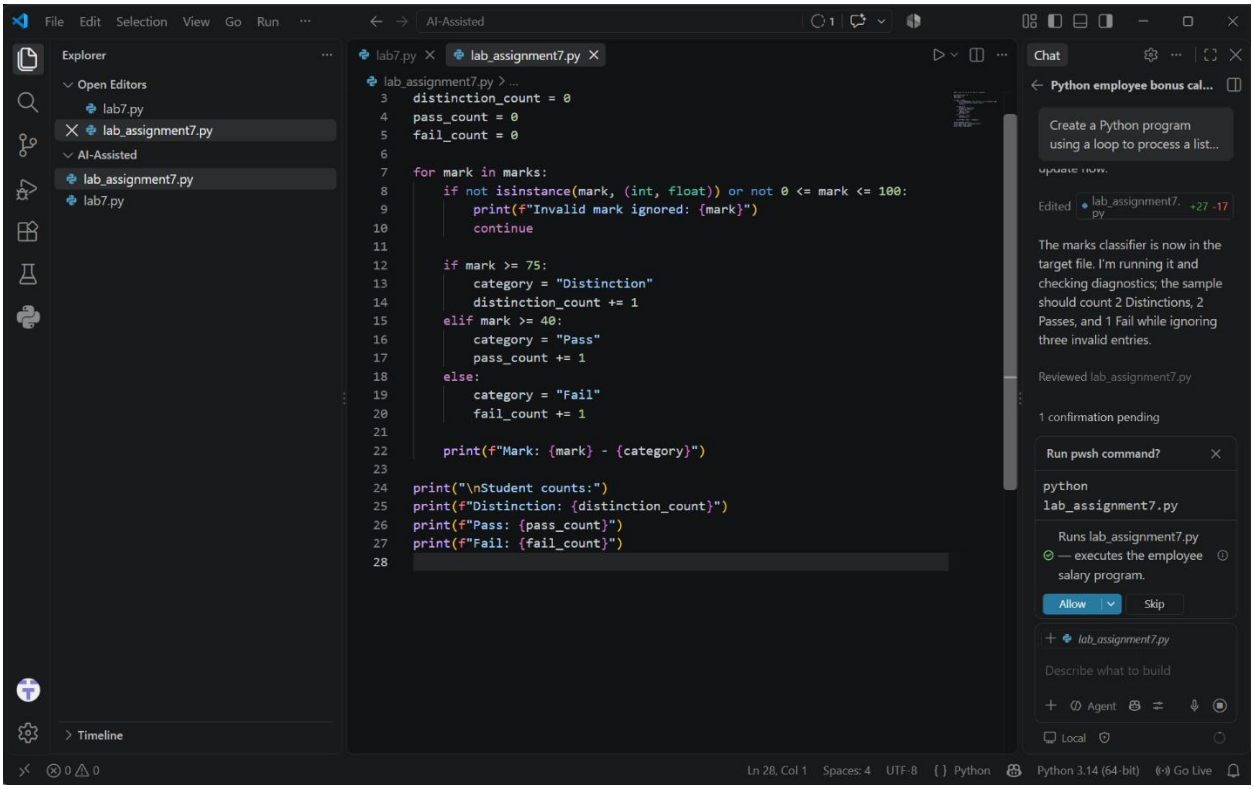
Scenario:
A teacher wants to analyze the marks of students using an AI coding assistant.

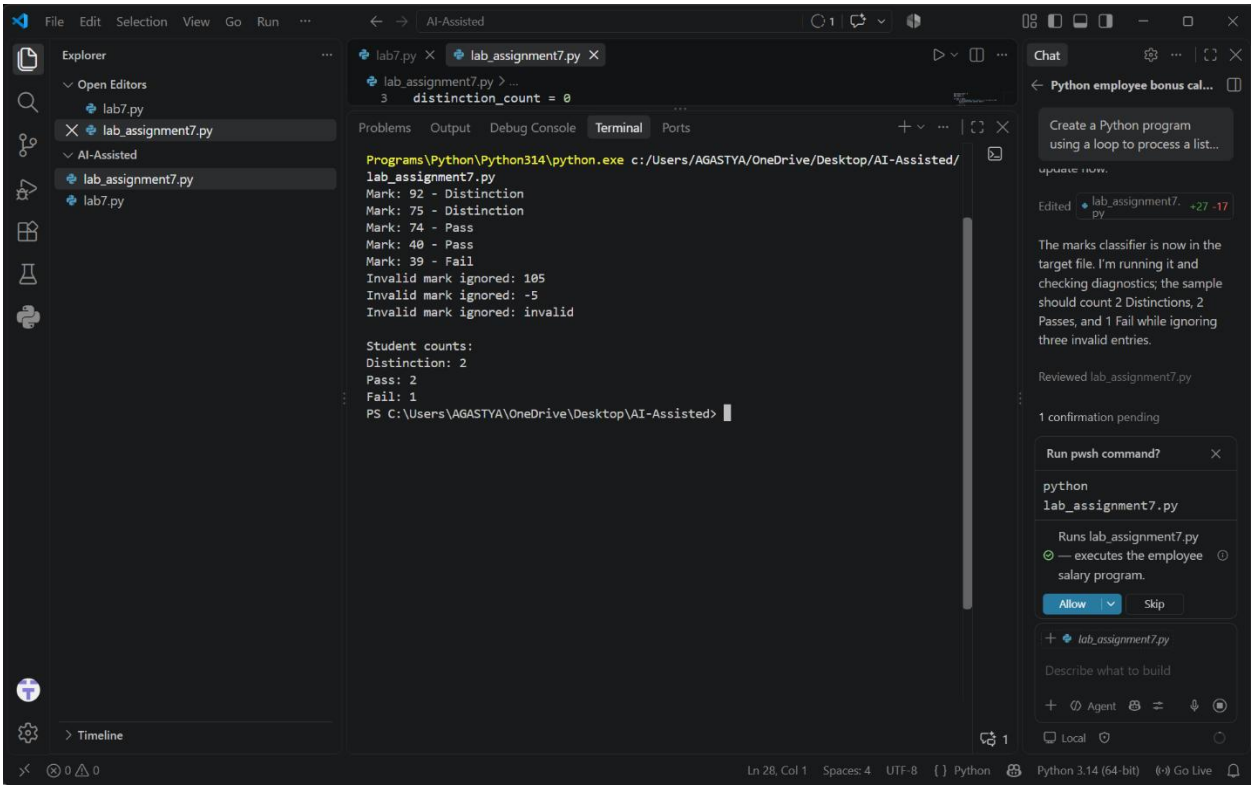
Task:
Using AI-assisted code completion, write a Python program that:

- Accepts a list of student marks.
- Uses a loop to process each mark.
- Displays:
    - "Distinction" for marks ≥ 75
    - "Pass" for marks ≥ 40 and < 75
    - "Fail" for marks < 40
- Ignores invalid marks outside 0–100.
- Calculates and displays the number of students in each category.
- Test the program with a suitable list of marks.

Prompt:
Create a Python program using a loop to process a list of student marks.
Display Distinction (≥75), Pass (40–74), or Fail (<40), and ignore invalid marks.
Count and display the number of students in each category.

Code & Output :


File Edit Selection View Go Run
AI-Assisted
Explorer
Open Editors
lab7.py
lab_assignment7.py
AI-Assisted
lab_assignment7.py
lab7.py
Timeline
lab7.py
lab_assignment7.py
lab_assignment7.py > ...
distinction_count = 0
pass_count = 0
fail_count = 0
for mark in marks:
if not isinstance(mark, (int, float)) or not 0 <= mark <= 100:
print(f"Invalid mark ignored: {mark}")
continue
if mark >= 75:
category = "Distinction"
distinction_count += 1
elif mark >= 40:
category = "Pass"
pass_count += 1
else:
category = "Fail"
fail_count += 1
print(f"Mark: {mark} - {category}")
print("\nStudent counts:")
print(f"Distinction: {distinction_count}")
print(f"Pass: {pass_count}")
print(f"Fail: {fail_count}")
Chat
Python employee bonus cal...
Create a Python program using a loop to process a list...
Edited lab_assignment7.py +27 -17
The marks classifier is now in the target file. I'm running it and checking diagnostics; the sample should count 2 Distinctions, 2 Passes, and 1 Fail while ignoring three invalid entries.
Reviewed lab_assignment7.py
1 confirmation pending
Run pwsh command?
python lab_assignment7.py
Runs lab_assignment7.py — executes the employee salary program.
Allow
Skip
lab_assignment7.py
Describe what to build
Agent
Local
Ln 28, Col 1 Spaces: 4 UTF-8 { } Python Python 3.14 (64-bit) Go Live



Observation:
Learned how to use loops and conditional statements to classify and count student marks.

Q3. Class + Loop – Product Inventory

Scenario:
A small store wants to maintain product inventory using a Python program generated with AI assistance.

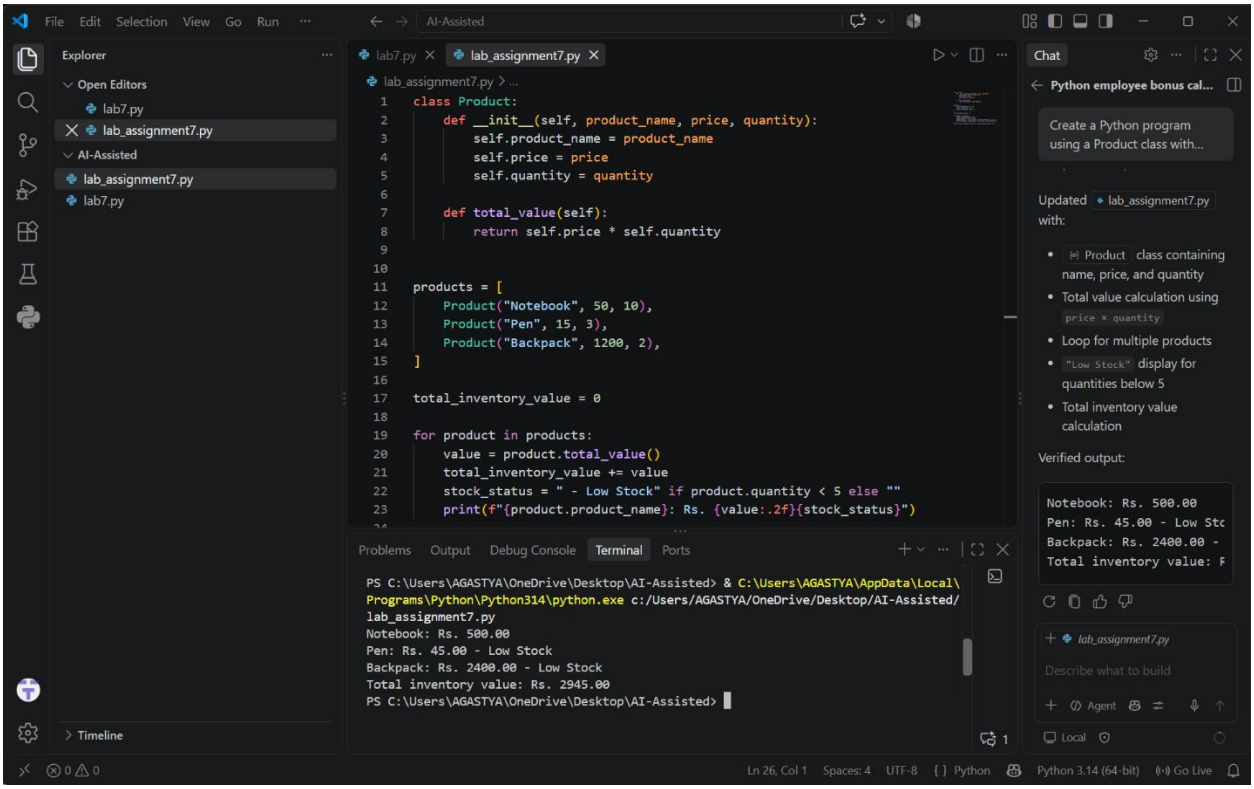
Task:
Using AI-assisted code completion, create a class Product that:

- **Stores product name, price, and quantity.**
- **Has a method total_value() that calculates price × quantity.**
- **Create a list of multiple Product objects.**
- **Use a loop to calculate the total inventory value.**
- **Display the products whose quantity is less than 5 as "Low Stock".**

- **Test the program with at least three products.**

Prompt:
Create a Python program using a Product class with product name, price, and quantity. Calculate total value using price × quantity and use a loop to process multiple products. Display products with quantity less than 5 as "Low Stock" and calculate total inventory value**.**

Code & Output :



Observation:
Learned to use classes and loops to manage product details and calculate total inventory value**.**

Q4. Class + Loop + Nested Conditions – Electricity Bill

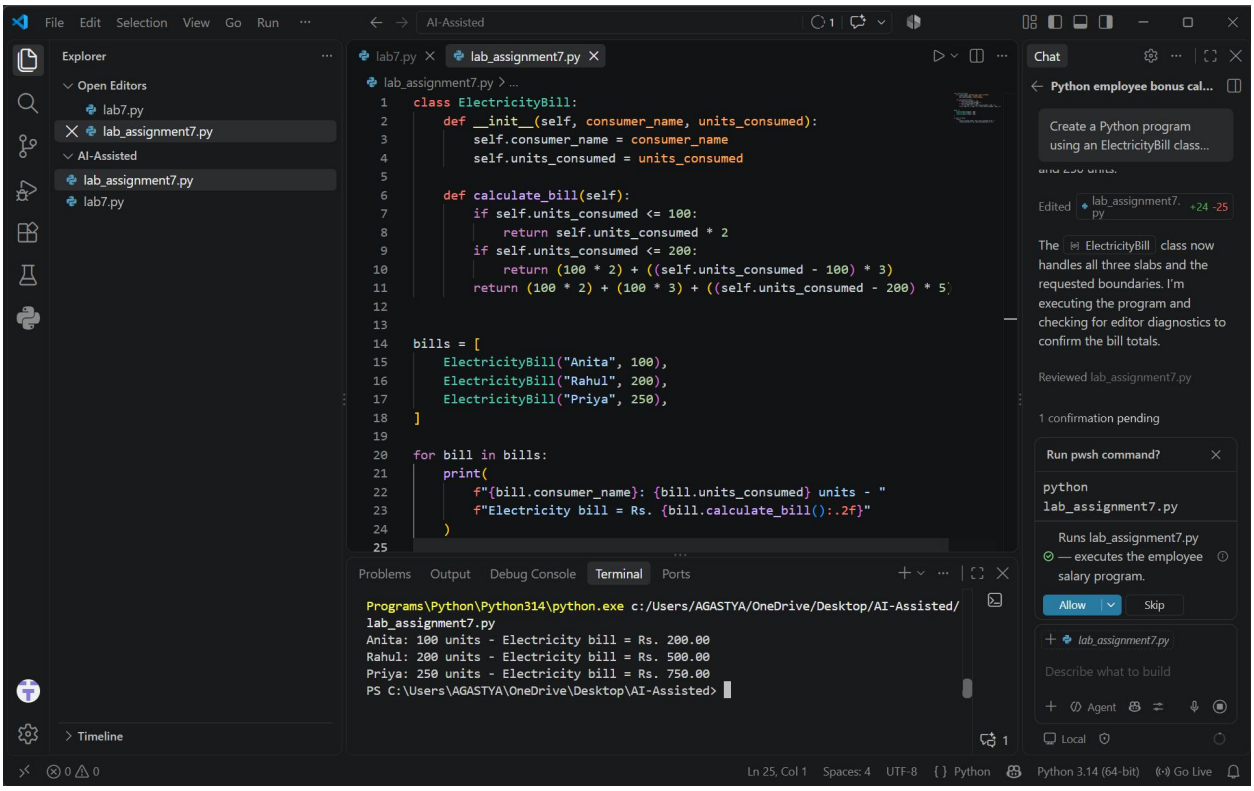
- **Scenario:**
  An electricity board wants to calculate bills based on units consumed.
- **Task:**
  Using AI-assisted code completion, create a Python class ElectricityBill that:
- **Stores the consumer name and units consumed.**
- **Calculates the bill using:**
- **First 100 units → ₹2/unit**
- **101–200 units → ₹3/unit**
- **Above 200 units → ₹5/unit**

- **Uses appropriate loops and conditional statements where required.**
- **Displays the consumer name, units, and total bill.**
- **Create at least three objects and display their bills.**
- **Test the program with different unit values, including an edge value such as 100 or 200.**

Prompt:

Create a Python program using an Electricity Bill class with consumer name and units consumed. Calculate the bill using ₹2/unit for the first 100, ₹3/unit for 101–200, and ₹5/unit above 200.Create three objects, display their bills, and test different unit values including 100 and 200**.**

Code & output:



Observation:

Learned to combine classes, loops, and conditional statements to calculate electricity bills based on units consumed.